# Çağatay Gürsoy

caggursoy93@gmail.com Heidelberg, DE TR + DE
github.com/caggursoy in/cagatay.gursoy@caggursoy.bsky.social

## Profile

Electrical & Electronics Engineer, research data specialist and Neuroscientist with a strong background in machine learning, large-scale data analysis, and the development of robust, reproducible analytical workflows. Experienced in designing and implementing containerized applications and automated data pipelines in collaborative, interdisciplinary settings. Skilled in extracting actionable insights from complex datasets and communicating results to diverse audiences. Currently completing a Ph.D. in Psychology/Neuroscience (expected End of year 2026), with a focus on leveraging advanced computational tools to solve challenging real-world problems.

## Areas of Expertise

Neuroimaging & fMRI Experiment Design - Behavioural Experiment Design - Machine Learning Applications
Containerization & DevOps - Version Control - CI/CD - Data Pipeline Development (ETL) - Data Workflow Automation
Open Science Practices - Pre-registration - Research Data Management
Data Visualization - Statistical Analysis - Scientific Project Management - Analytical Problem Solving

## Professional Experience

**Research Data Specialist,** *Springer Nature* **Heidelberg, DE** *09/2025 - present*

- **Research Data Team - Content Innovation**:
  - Building synergies between Springer Nature and EMBO Press on AI-based research tools.
  - Experimenting on and developing cutting edge AI-tools that aids the modern research cycle.

**Doctoral Researcher,** *Zentralinstitut für Seelische Gesundheit (ZI)* **Mannheim, DE** *09/2019 - 09/2025*

- **Targeted Extinction of Drug Cues During Sleep (TEDDS):** On-site fMRI+Sleep EEG experiment enhancing precision in recording targeted cue reactivation with combined olfactory and visual cues.
  - Managed the entire project lifecycle from conception to publication; coded experiments with PsychoPy, enrolled participants, collected fMRI data, performed statistical analysis on behavioural and neuroimaging data - closely following ETL principles - and submitted the manuscript.
- **Multilevel Feature Learning:** Online experiment investigating model-based/model-free human learning upon objects with varying multi-level features.
  - Managed the entire project lifecycle from DevOps to publication; recruited participants via Prolific, coded experiment with Javascript/jsPsych, containerized the study with internal ZI-GitLab, collected behavioral data, performed statistical analysis - closely following ETL principles - and submitted the manuscript.
- **Breathing Apparception Training (BAT):** Breathing training for persons with anxiety disorder.
  - Joined the project after data collection. Handled data cleaning, statistical analysis and manuscript submission.

**Project Developer and Maintainer,** *IGOR-bioDGPs - ARIADNE* *10/2022 - present*

- **ARIADNE**: An open science navigator project helping Early Career Researchers find accurate research tools.
  - Created and maintained an online JupyterBook hosted on GitHub with a dynamic Cytoscape.js network visualizing resources and a 10-step research roadmap.
  - Published manuscript in Advances in Methods and Practices in Psychological Science.

## Education

**Ph.D. in Psychology** *Ruprecht-Karls-Universität Heidelberg* **Heidelberg, DE** *08.2023 - expected finish 09.2026*
**M.Sc. in Cognitive Sciences** *Middle East Technical University* **Ankara, TR** *02.2017 - 07.2019*
**B.Sc. in Electrical and Electronics Engineering** *Bilkent University* **Ankara, TR** *09.2011- 01.2016*

## Skills

- **Research & Data Analysis:** Statistical Analysis, fMRI Experiment Design, Behavioral Experiment Design, Pre-registration, Analytical Problem Solving
- **Programming:** Python, PsychoPy, MATLAB, R, Javascript, JsPsych, SPM

- **Software & Tools:** Version Control (Git), Containerization, CI/CD, JupyterBook, Cytoscape.js
- **Open Science:** Research Data Management, FAIR Principles, Open Access Publishing, Reproducible Workflows
- **Soft Skills:** Project Management, Supervision, Teamwork, Cross-functional Collaboration, Problem-Solving, Adaptability

## Languages

- **Turkish** [Native]
- **English** [Professional]
- **German** [B2]

## Selected Publications & Presentations

- H. Hartmann, **Ç. Gürsoy**, A. Lischke, et al., "Ariadne: A scientific navigator to find your way through the resource labyrinth of psychological sciences," *Advances in Methods and Practices in Psychological Science*, vol. 8, no. 1, 2025. https://doi.org/10.1177/25152459241297674
- L. Puhlmann, A. Koppold, G. Feld, T. Lonsdorf, K. Hilger, S. Vogel, **Ç. Gürsoy**, A. Kastrinogiannis, L. Kulke, A. Lischke, A. Müller-Alcazar, J. Packheiser, M. Sperl, Y. Yang, L. Bechtold, S. Ocklenburg & H. Hartmann, "Sustainable neuroscience through open science," *Nature human behaviour*, 10(4), 639–642, 2026. https://doi.org/10.1038/s41562-026-02426-3
- **Ç. Gürsoy**, "ARIADNE: A scientific navigator for early career researchers," *50th Annual Conference on "Psychology and the Brain*, Würzburg, 2025. *Oral Presentation*
- **Ç. Gürsoy** and G. B. Feld "Validation of an fmri-based olfactory cue reactivity task," *2023 International Conference on Learning and Memory*, Huntington Beach, CA, USA, 2023. *Oral Presentation*

## Workshops Given

- **Introduction to fMRI experiment scripting with PsychoPy**
  Prepared and delivered an introductory course about constructing fMRI experiments with PsychoPy at the Institute for Psychosocial Prevention, Centre for Psychosocial Medicine, University of Heidelberg.
- **fMRI Analysis Workgroup**
  Initiated a workgroup of fMRI data preprocessing and analysis within the Dept. of Clinical Psychology, ZI.

## Extracurricular

- **ZI-ECRs Network**
  Founded and led the Early Career Researchers (ECR) network at ZI, organizing regular networking events, lectures, and workshop series to foster collaboration and professional development.
- **Knowledge-Transfer Curriculum**
  Contributed to the design and implementation of a knowledge-transfer curriculum for ECRs, supporting skill development and interdisciplinary exchange.
- **ZI-ECRs Internationals**
  Established and coordinated a support group for international ECRs, facilitating their integration into the institute and local community.
- **Deep Face Perceiver – Digital Artwork**
  Created an AI-generated digital art piece as an extension of my MSc research, exhibited at Figure 1.a. 2021.

## References

- **Graham Smith**
  Director Research Data Innovation, Content Innovation, Springer Nature
- **Prof. Dr. Peter Kirsch**
  Department Head, Department of Clinical Psychology, ZI, Mannheim
- **PD Dr. rer. nat. Gordon Feld**
  Head of Research Group, RG Psychology and Neurobiology of Sleep and Memory, ZI, Mannheim